Functional Software Design (FSD) Report

Auto Ticket Classification using ServiceNow Flow Designer

Project Title:	Auto Ticket Classification using Flow Designer
Document Type:	Functional Software Design (FSD) Document
System Architecture:	ServiceNow Platform / Flow Designer (No-Code Workflow)
Team Id:	SWTID-2026-6328

1. Introduction

This Functional Software Design (FSD) document outlines the architectural specifications, configuration steps, and operational workflows for automated incident classification within a school IT helpdesk ecosystem.

1.1 Project Title

Auto Ticket Classification using ServiceNow Flow Designer

1.2 Team Members & Roles

- **ServiceNow Administrator / Architect:** Configures custom tables, field dictionaries, update sets, and access controls.
- **Workflow & Automation Engineer:** Designs, tests, and deploys trigger conditions, flow logic, and email notifications in Flow Designer.
- **IT Operations Lead / QA Specialist:** Validates ticket routing accuracy, dependent drop-down choices, and end-to-end integration testing.

2. Project Overview & Business Case

2.1 Problem Statement

The school IT helpdesk processes high daily volumes of incident requests submitted by students and teachers. Issues span network connectivity (Wi-Fi), hardware failures (projectors), user authentication (forgot passwords), and system responsiveness (slow computers). Manual ticket evaluation is highly time-consuming, prone to human error, and fails to scale as support volume grows.

2.2 Purpose & Business Objectives

- **Automated Classification:** Instantly categorize tickets upon submission without requiring human intervention.
- **Operational Efficiency:** Reduce ticket triage time and relieve manual burden on IT helpdesk agents.
- **Improved Routing:** Ensure tickets reach responsible support groups immediately with standardized metadata.
- **No-Code Maintainability:** Leverage native ServiceNow Flow Designer capability for scalable, script-free maintenance.

2.3 Core Features

- **Dynamic Field Dependencies:** Dependent drop-down logic restricting subcategory selections based on the chosen category.
- **Keyword-Driven Parsing:** Automatic evaluation of Short Description text against target keywords (WiFi, Projector, Password, Slow).
- **Automated Email Acknowledgments:** Instant notification emails sent directly to callers upon ticket creation.
- **Update Set Portability:** Complete packaging of customization records for smooth deployment across platform environments.

3. Architecture & Backend Data Structure

3.1 ServiceNow Data Schema (`u_incident_workflow`)

All incident workflow records are maintained in a custom ServiceNow table named Incident WorkFlow (`u_incident_workflow`). The table structure and data types are defined below:

Field Label	Data Type	Reference / Choice Options
Number	Auto Number	INC00500 (Prefix: INC, Digits: 5, Number: 500)
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice (Dependent)	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	Free text issue summary
Description	String	Detailed issue information
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group

Assigned to	Reference	sys_user

3.2 Choice Dependency Matrix

To preserve data integrity, the Subcategory field relies on Category as its dependent field:

Category Value	Subcategory Label	Dependent Choice Value
Network	Wi-Fi	network
Hardware	Projector	hardware
Access	Forgot Password	access
Performance	Slow Computer	performance

4. Setup & Implementation Guide

4.1 Prerequisites

- **ServiceNow Instance:** Vancouver/Washington release or higher with System Administrator role.
- **Plugins Enabled:** Process Automation - Flow Designer & Core Incident Management.

4.2 Update Set Initialization

- **Step 1:** Navigate to All -> System Update Sets -> Local Update Sets.
- **Step 2:** Click New. Set Name = 'Project Update Set', State = 'In progress', Application = 'Global'. Save the form.
- **Step 3:** Click Related Link 'Make This My Current Set' to track all subsequent dictionary and workflow changes.

4.3 Custom Table & Form Configuration

- **Table Creation:** Navigate to System Definition -> Tables. Click New. Label = 'Incident WorkFlow', Name = 'u_incident_workflow'.
- **Auto Numbering:** In Controls tab, enable 'Auto-number'. Set Prefix = 'INC', Number = 500, Digits = 5.
- **Form Designer:** Open context menu -> Configure -> Form Design. Add fields per the schema matrix.
- **Dependency Setup:** Right-click Subcategory -> Configure Dictionary. Under Dependent Field section, enable 'Use dependent field' = True, set 'Dependent on field' = 'Category'. Set choice backend mappings.

5. Automation Logic (Flow Designer)

5.1 Flow Specification: `Auto Classify School IT Tickets`

- **Trigger:** Record Created on `Incident WorkFlow` (`u_incident_workflow`) table where `Category` is Empty.
- **Flow Name:** Auto Classify School IT Tickets (Application Scope: Global).

5.2 Logic & Rule Execution Matrix

Condition Block	Keyword Rule (Short Description)	Assigned Category	Assigned Subcategory
If Block 1	Contains 'Wi-Fi' OR 'Network'	Network	Wi-Fi
Else If Block 2	Contains 'Projector' OR 'Hardware'	Hardware	Projector
Else If Block 3	Contains 'Forgot password' OR 'Password' OR 'Login'	Access	Forgot Password
Else If Block 4	Contains 'Slow computer' OR 'Slow' OR 'Hanging'	Performance	Slow Computer

5.3 Notification Action

Following successful categorization, Flow Designer triggers a 'Send Email' action with the following parameters:

- **To:** Data Pill: Trigger -> Incident WorkFlow Record -> Caller -> Email
- **Subject:** Your Request for the issue has been submitted.
- **Body:** Hi greetings from xyz college!!! We have received your request and will solve it in a short time period. Thank you.

6. Testing, Validation & Verification

6.1 Test Scenarios & Execution Results

Scenario ID	Input Short Description	Expected Category	Expected Subcategory	Notification Status
TS-001	WiFi not working in library	Network	Wi-Fi	Email Sent (SMTP Send-Ready)
TS-002	Projector not	Hardware	Projector	Email Sent

	turning on	(SMTP Send-Ready)

6.2 Data Integrity Verification

- **Auto-Numbering:** Verified sequential ticket generation (e.g., INC00511, INC00512) without duplicate records.
- **Notification Logs:** Verified under System Logs -> Emails. Email preview confirms correct recipient resolution.

7. Deployment & Export

- **Update Set Finalization:** Navigate to Local Update Sets -> Open 'Project Update Set' -> Change State from 'In progress' to 'Complete'. Save form.
- **XML Export:** Click 'Export to XML' under Related Links. Download `sys_remote_update_set_a7b40de6fc590ceaad36c.xml` for distribution across development, testing, and production instances.

8. Known Issues & Future Enhancements

8.1 Known Issues / Edge Cases

- **Unmatched Keyword Fallback:** If a ticket short description lacks predefined keywords, category remains unassigned and requires tier-1 agent triage.

8.2 Future Enhancements

- **Predictive Intelligence:** Integrate Machine Learning / AI Classification for complex, multi-sentence issue descriptions.
- **Automated Group Assignment:** Extend flow logic to dynamically populate 'Assigned Group' based on Subcategory.
- **SLA Tracking:** Incorporate Service Level Agreement (SLA) timers based on calculated priority.

9. Conclusion

The Auto Ticket Classification using Flow Designer project successfully delivers a reliable, no-code automation solution for school IT helpdesk operations. By combining structured backend schema definitions, dependent choice logic, keyword parsing, and automated notification workflows, the platform eliminates triage bottlenecks and establishes a scalable foundation for enterprise IT service management.